

ARFW501

Group D – Final Presentation

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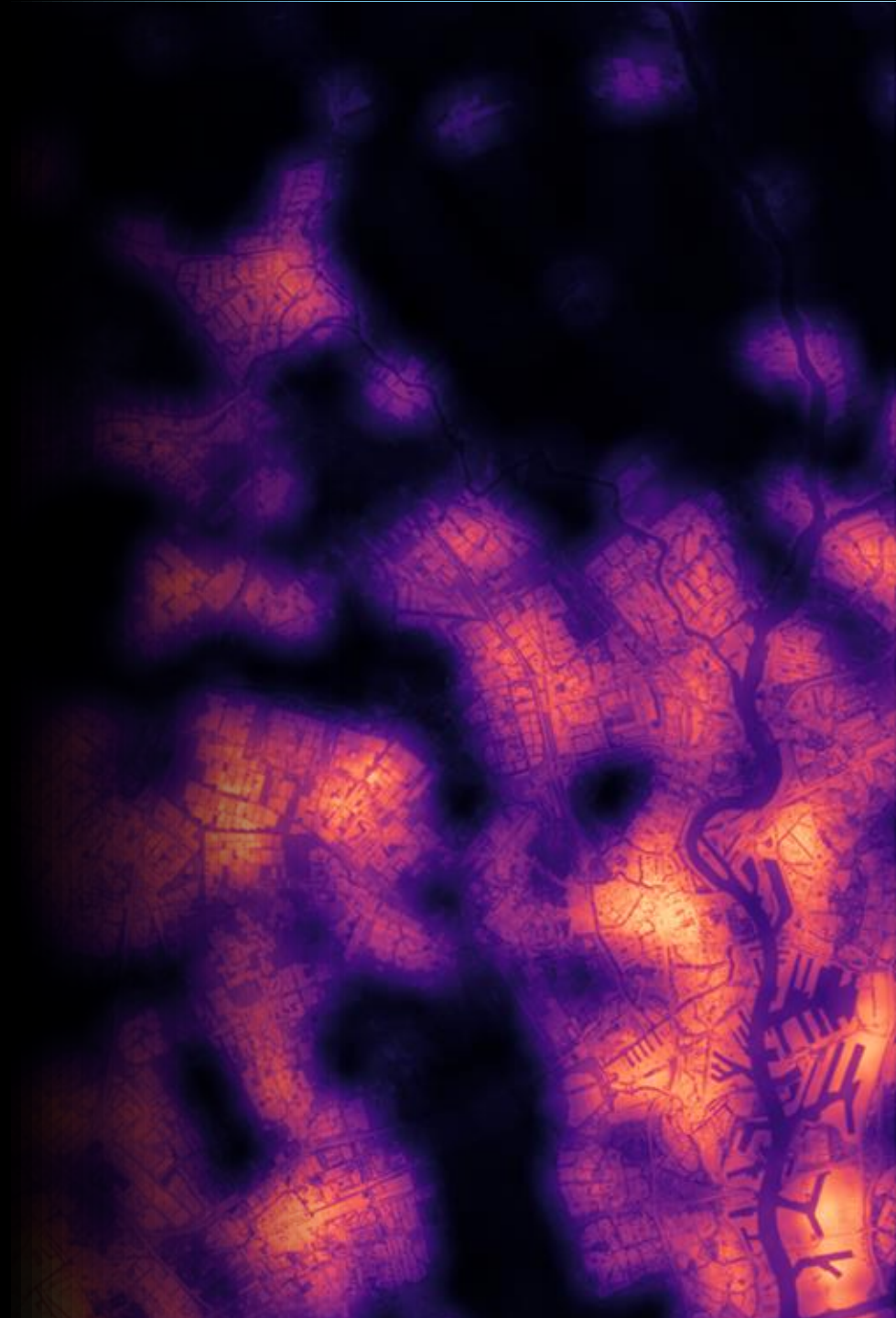
The Urban Challenge

Climate change and rapid urbanization are reshaping our local environments
causing UHI effect

Defining the UHI Effect and its primary drivers

Cities become "islands" of trapped heat compared to their rural counterparts. This phenomenon is driven by the physical properties of urban surfaces, human density and activity.

- **Absorption:** Dark surfaces like asphalt soak up solar radiation.
- **Retention:** Building materials store thermal energy throughout the day.
- **Emission:** Heat is released slowly at night, preventing cooling.



Research Framework

Bridging urban design with spatial data science for UHI mitigation

“How can multi-scale spatial analysis be used to identify locations where nature-based solutions can most effectively mitigate urban heat island effects and what nature-based solutions are suitable on street level?”

- Core Research Question

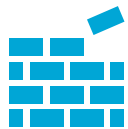
Literature Review

Parameters considered in research

basis for data requirements and conceptual framework



○ LCZ (Local Climate Zones)



○ Land Surface Temperature

○ NDVI

○ Wind speed

○ UHI



Nomenclature

Af	Tropical rainforest climate
AH	Anthropogenic heat
a_s	Albedo of the building surfaces
Aw	Tropical wet and dry or savanna climate
b	Empirical Clapp and Hornberger constant
BEM	Building energy models
BEU	Building electricity usage (kWh)
BLUP	Best linear unbiased prediction
Bsk	Cold semi-arid climate
Bwh	Hot desert climate
c	Substrate water content
C	Concentration
Cfa	Humid subtropical climate
Cfb	Temperate oceanic climate
CFD	Computational fluid dynamics
C_f	Drag effect for canopy
CHTC	Convective heat transfer coefficient (W/(m ² ·K))
Csa	Hot-summer Mediterranean climate
CV-RMSE	Coefficient of the variation of the RMSE (%)
Cwa	Humid subtropical climate
d	Index of agreement
D_{ave}	Average deviation value
Dfa	Hot-summer humid continental climate
Dfb	Warm-summer humid continental climate
DLNM	Distributed lag non-linear models
D_{max}	Maximum deviation value
Dwa	Hot-summer humid continental climate
EBM	Energy balance models
FF	Flow field
F_i	Reduction of momentum
F_k	Increase in turbulence
F_ε	Turbulence energy dissipation
GnPR	Green plot ratio
GR	Global solar radiation (W/m ²)
HVAC	Heating, Ventilation, and Air-Conditioning
HF	Heat flux (W/m ²)
HI	Heat index
HSI	Heat stress index
k	Turbulence energy
LAD	Leaf angle distribution
LAI	Leaf area index
LE	Latent heat flux
MAE	Mean absolute error
MBE	Mean bias error
MMT	Minimum mortality temperature (°C)

MODIS	Moderate Resolution Imaging Spectroradiometer
MRM	Mortality risk model
NDVI	Normalized Difference Vegetation Index
PAR	Photosynthetically active radiation (W/m ²)
PET	Physiologically equivalent temperature (°C)
PIV	Particle Image Velocimetry
PPI	Pores per inch
Q	Radiation
r	Pearson correlation coefficient
R²	Coefficient of determination
RH	Relative humidity (%)
RH_{in}	Indoor relative humidity (%)
RL	Longwave radiation (W/m ²)
RMSE	Root mean square error
RR	Relative risk
SH	Specific humidity(g/kg)
SVF	Sky view factor
T_a	Ambient temperature (°C)
T_g	Globe temperature (°C)
T_{in}	Indoor temperature (°C)
T_{ls}	Leaf surface temperature (°C)
T_{mrt}	Mean radiant temperature (°C)
T_{op}	Operative temperature (°C)
T_s	Surface temperature(°C)
TSF	Total site factor
T_{sin}	Interior surface temperature (°C)
TVF	Tree view factor
TMY	Typical meteorological year
u	Air speed
UCA	Urban canyon albedo
UCM	Urban Canopy Model
UGI	Urban green infrastructure
UHI	Urban heat island
UTCI	Universal Thermal Climate Index (°C)
UV	Ultraviolet
U-value	Thermal conductivity (W/m·K)
VPD	Vapour pressure deficit
WBGT	Wet-bulb globe temperature (°C)
WD	Wind direction (°)
WRF	Weather research forecast
WS	Wind speed (m/s)
z	Height (m)
η	The fraction of the area covered with vegetations
ε	Energy dissipation rate
λ_s	Thermal conductivity of the substrate layer
ρ_sc_s	Volumetric heat capacity
Ψ	Moisture potential

Comparative Global Analysis

Architectural & Climatic Contrast

Rotterdam, Temperate maritime hub



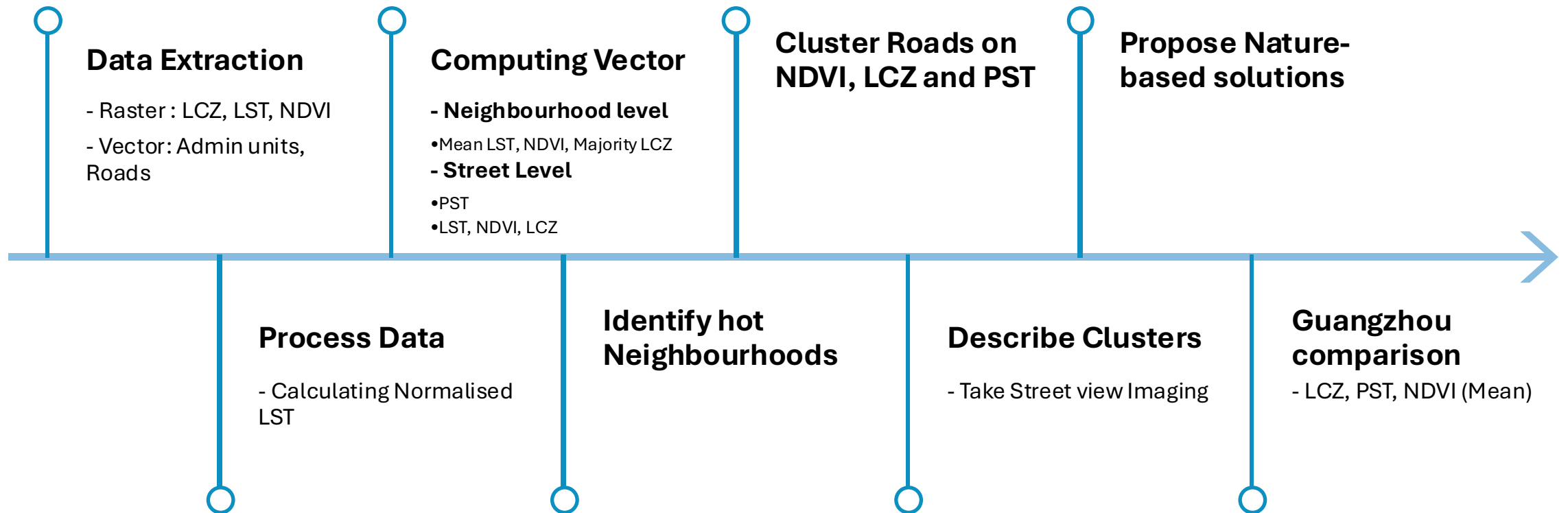
A northern European port city defined by modern architecture and a temperate maritime climate. Focus on water management and port-side cooling.

Guangzhou: Subtropical Density



A subtropical economic powerhouse in southern China. A mix of traditional urban villages and soaring contemporary skyscrapers with Pearl River influence.

Methodology

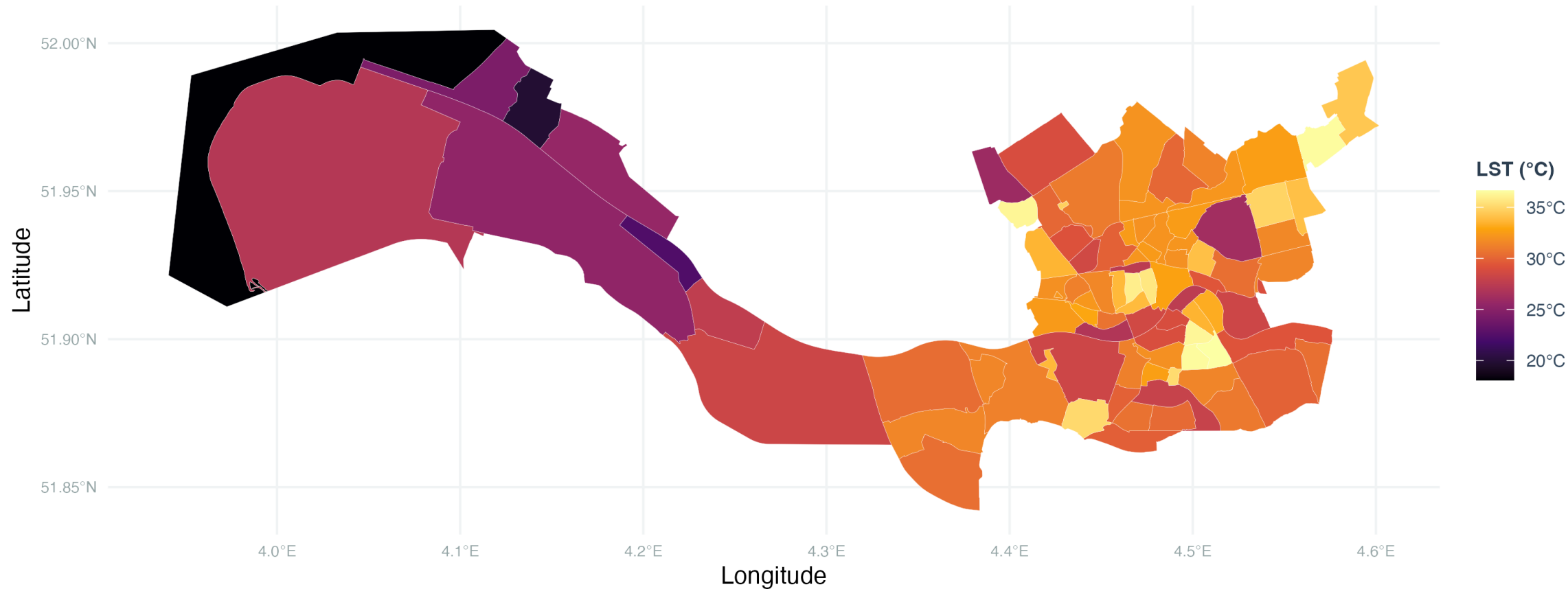


02

Intermediate Large-scale output

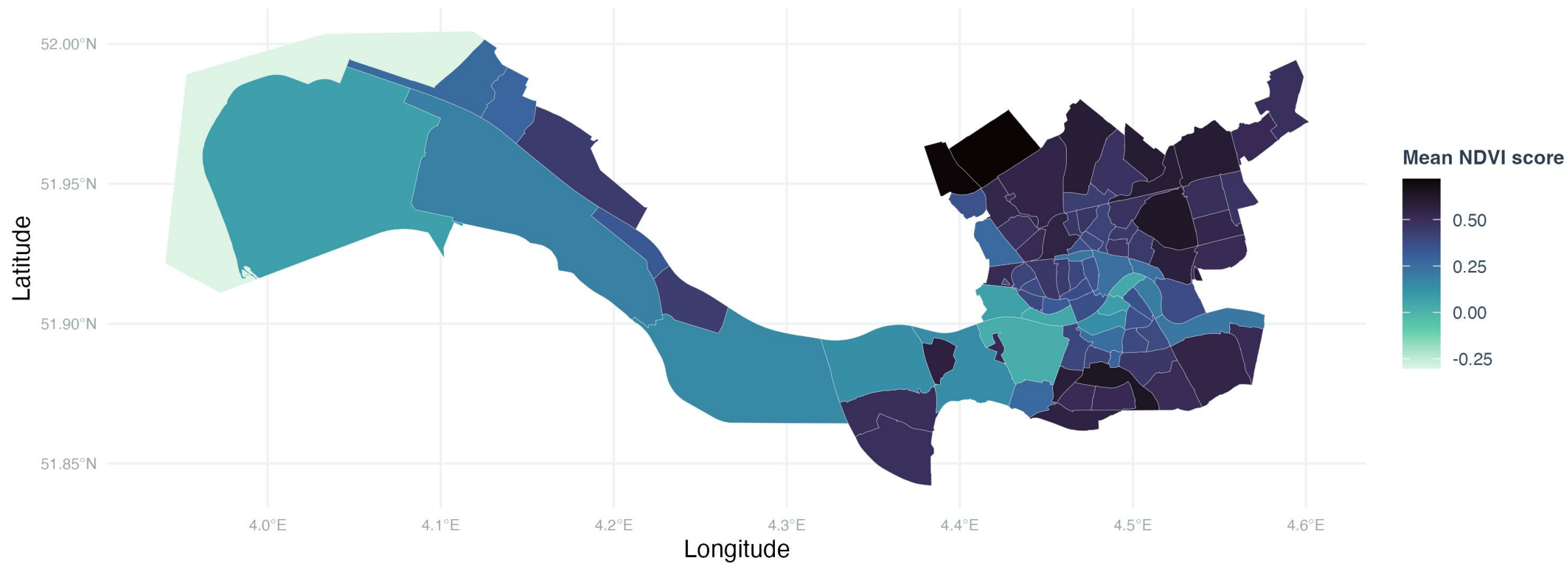
Rotterdam Land Surface Temperature Surface (LST)

Aggregated thermal baseline map at neighborhood scale



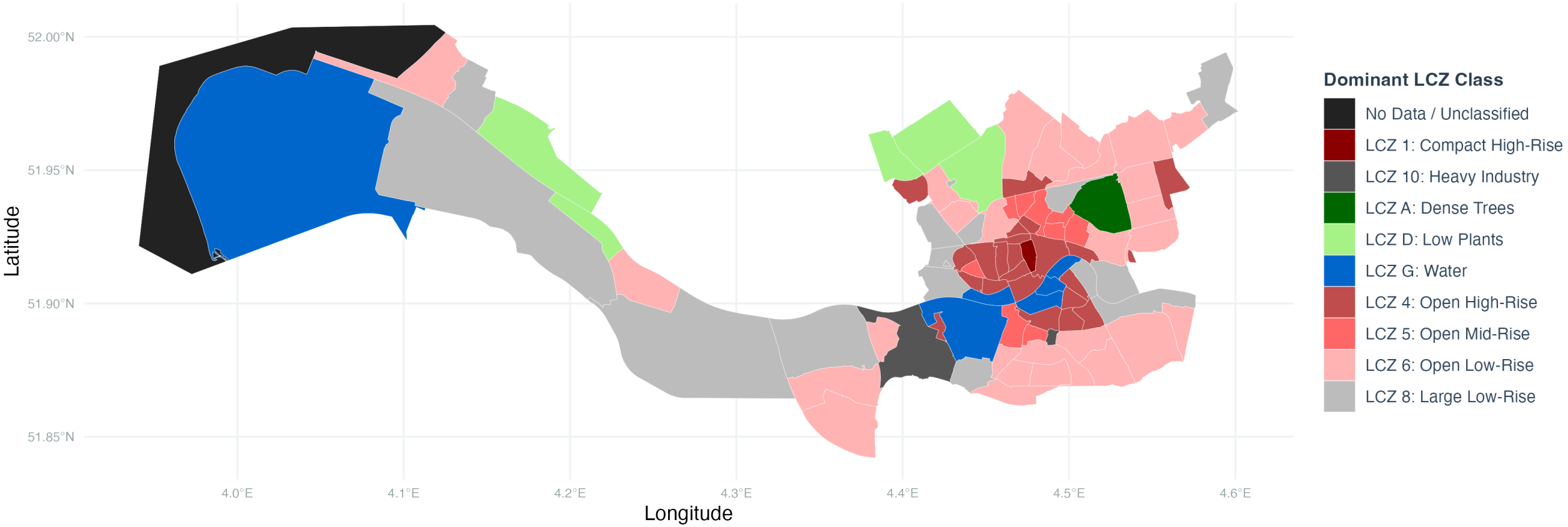
Rotterdam NDVI Vegetation Index Distribution

Mean Normalized Difference Vegetation Index (NDVI) values



Rotterdam Local Climate Zone Layouts (LCZ)

Standardized WMO classification scheme for urban microclimate typologies



03

Rotterdam: Street-wise Clustering

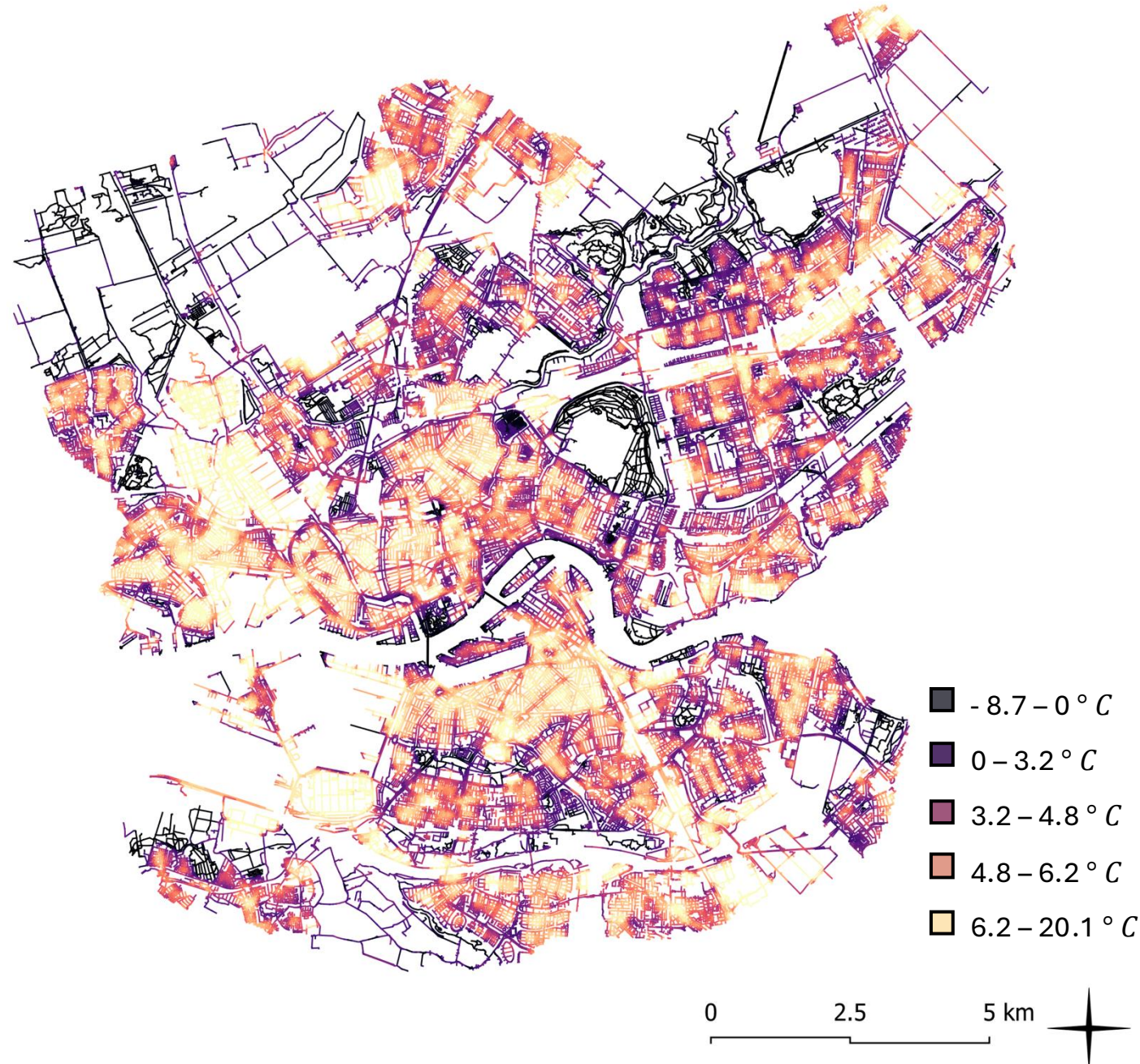
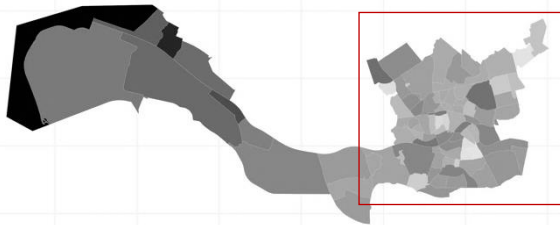
Street-wise LST

Superimposition of LST

Thermal hotspots occur at a much finer scale than neighbourhood boundaries

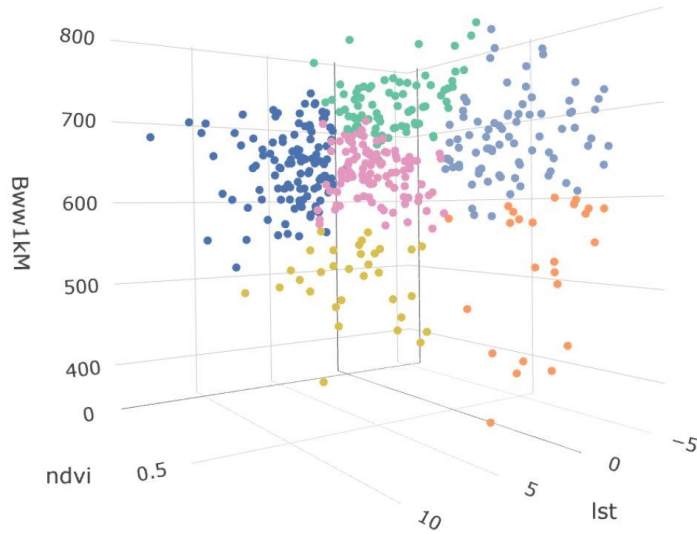
Street-level mapping enables targeted climate adaptation planning

Hot streets require additional context before selecting interventions

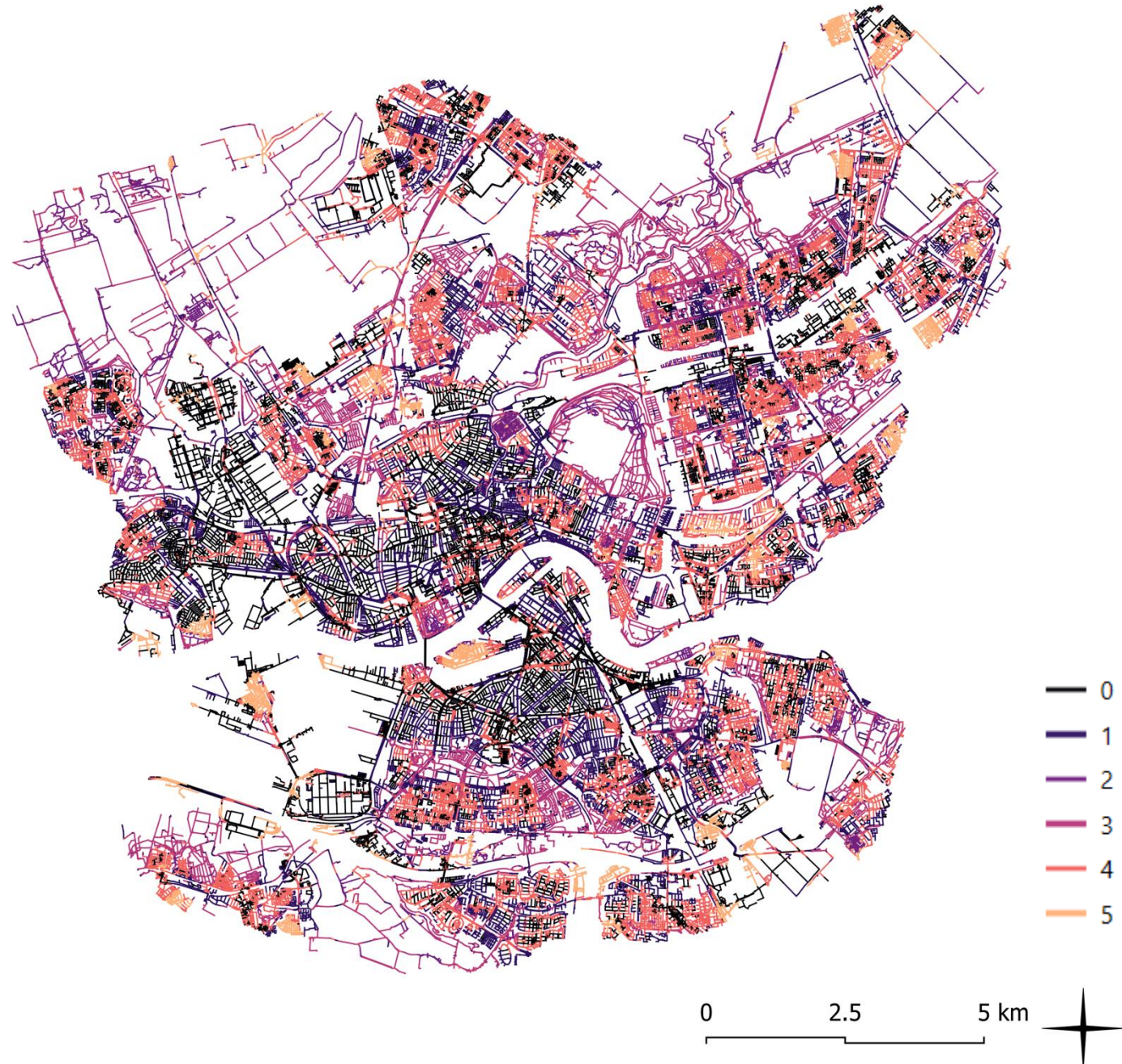


Street-wise clustering

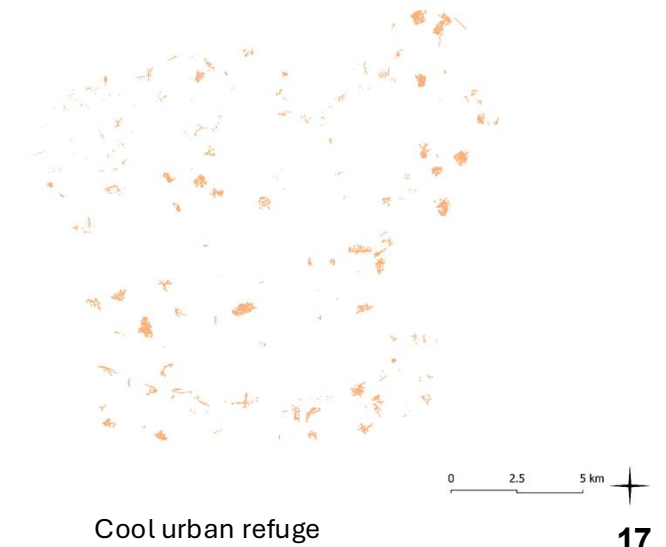
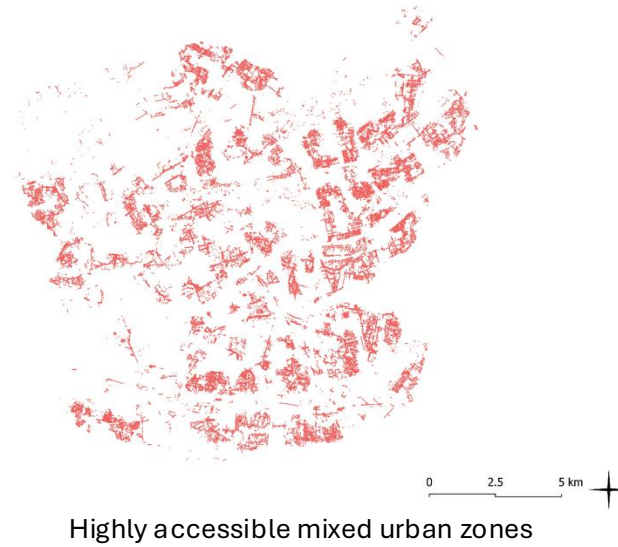
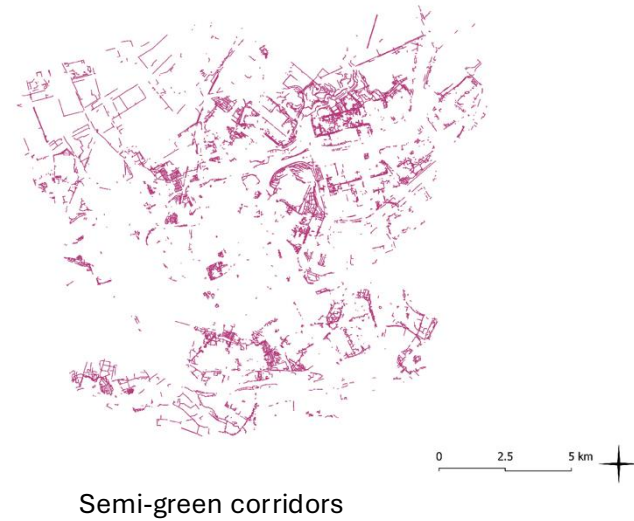
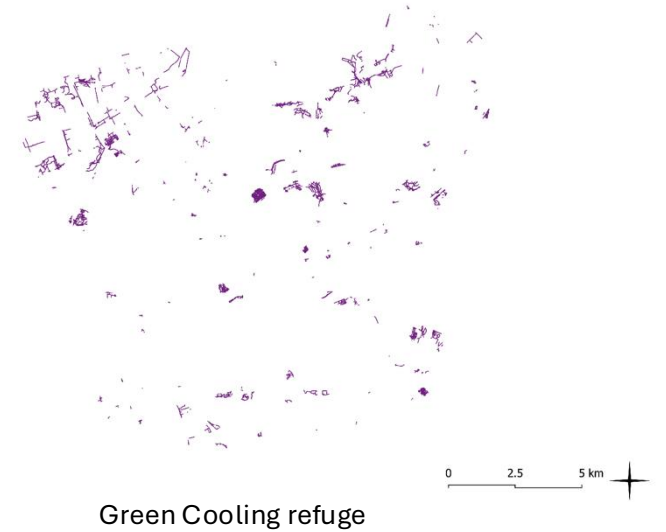
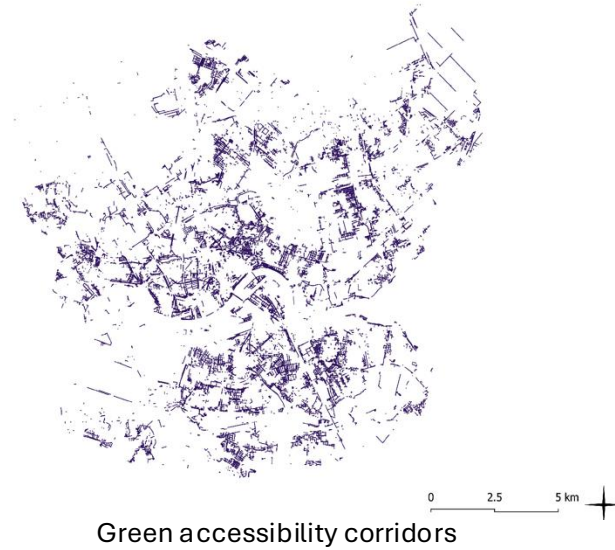
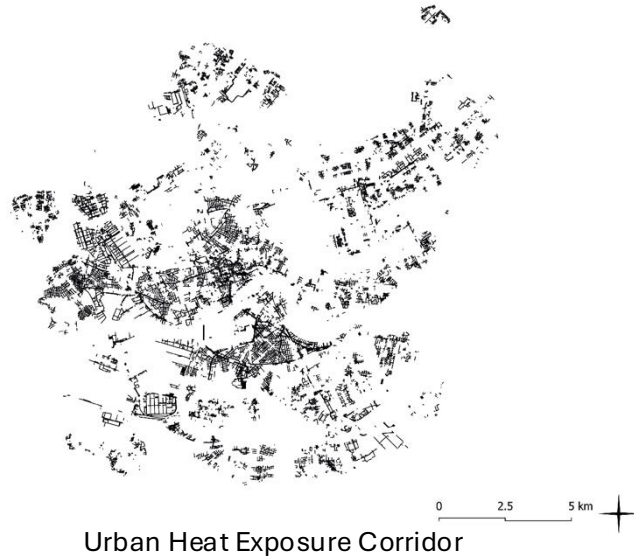
relating LST, NDVI and PST



- 0. High LST / Low NDVI / Elevated PST
- 1. Medium LST / Medium NDVI / High PST
- 2. Low LST / High NDVI / Low PST
- 3. Low LST / Low NDVI / High PST
- 4. Medium LST / Medium NDVI / High PST
- 5. Medium LST / Medium NDVI / Low PST



Cluster-types



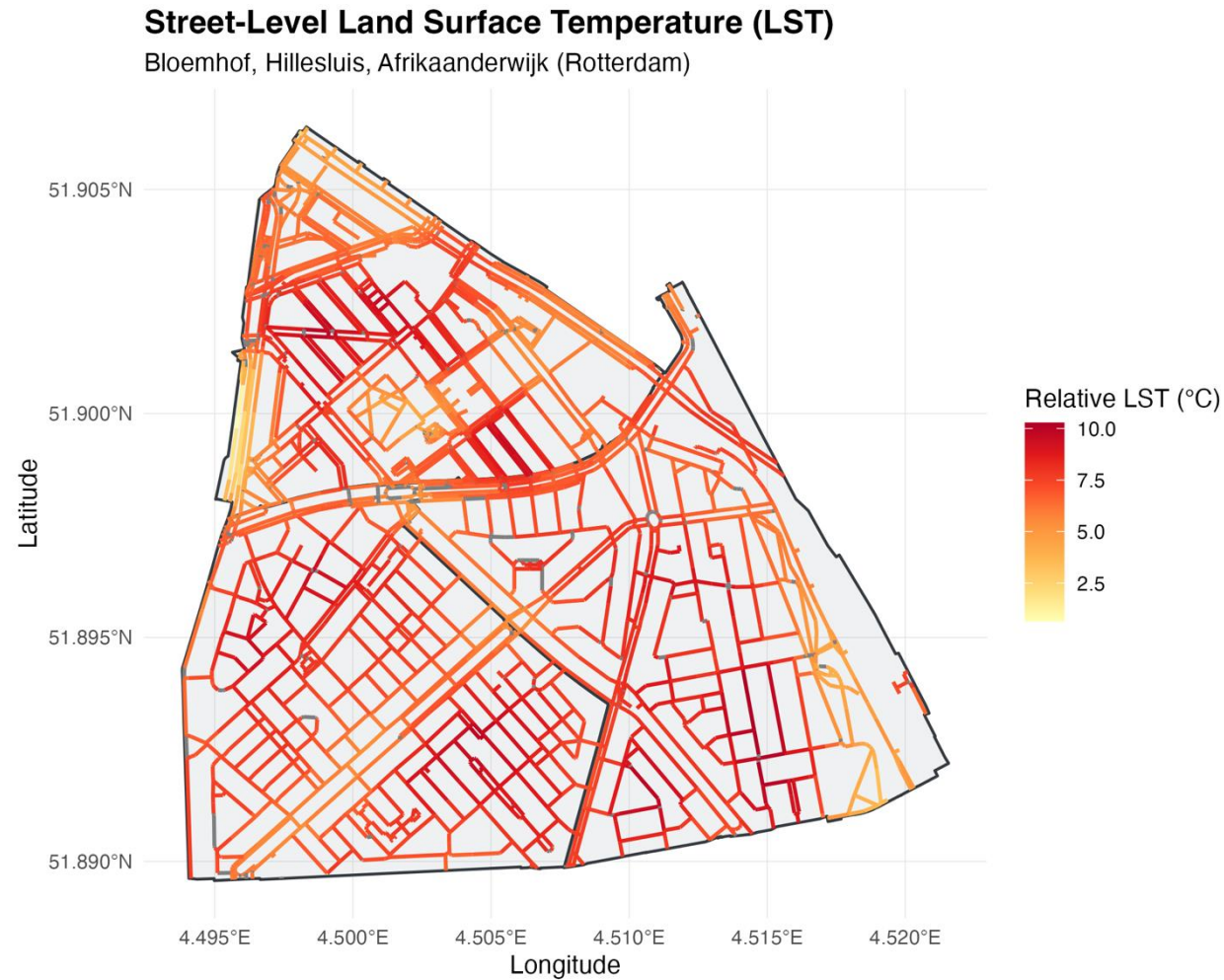
Spaanse polder



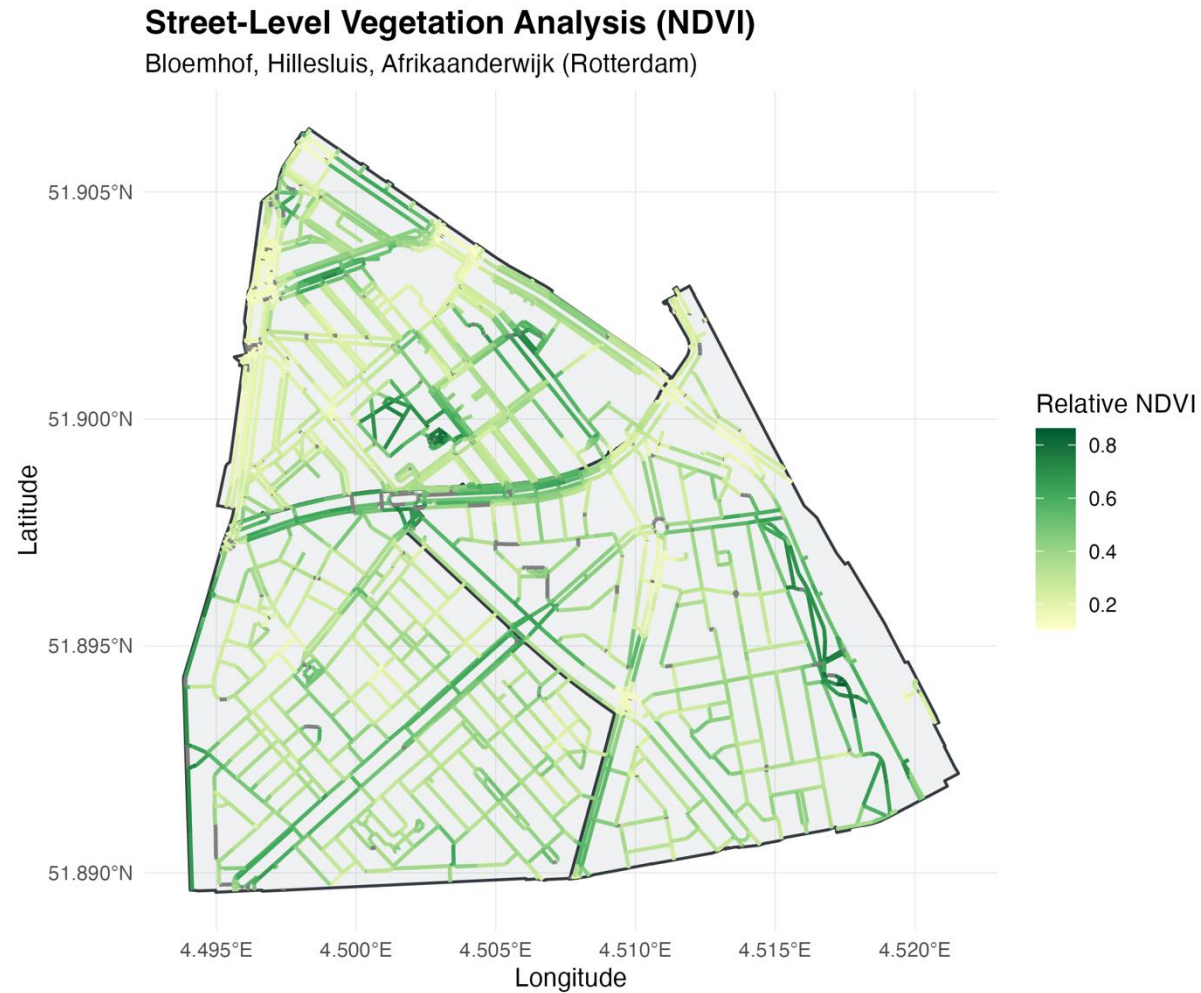
CS Kwartier



Bloemhof, Hillesluis and Afrikaanderwijk



Bloemhof, Hillesluis and Afrikaanderwijk





Bloemhof, Hillesluis and Afrikaanderwijk

Cluster 0

Highest temp streets



04

Rotterdam: Street-wise Nature-based Solutions

**For three hottest neighbourhoods
Bloemhof, Hillesluis and Afrikaanderwijk**



4.1 Extensive green roof

Extensive green roofs are lightweight systems that consist of a thin substrate layer with shallow-rooted, low growing, and often readily spreading vegetation. Typical groups of vegetation for extensive green roofs include sedums, herbs, wildflowers, grasses, and mosses, since they are relatively hardy and can often survive in low-nutrient conditions. Once established, extensive green roofs are characterized by their minimal maintenance and management requirements. However, they are often only accessible for maintenance purposes and not open to the public.

Compared to typical grey roofs, extensive green roofs can offer benefits like localized air temperature and pollution reduction, and contribute to water management. These benefits, however, tend to be less extensive than those associated with their more complex and expensive counterparts - intensive green roofs.



4.2 Intensive green roof

Intensive green roofs are heavier greening systems characterized by a thicker growing medium with more varied vegetation types (compared to extensive green roofs). Common plants used for intensive green roofs include a variety of smaller trees, shrubs, and perennials. Depending on design, intensive green roofs provide many benefits like stormwater storage, reduction of air and water pollution, reduction of localized air temperature, and biodiversity enhancement. They are commonly found on residential buildings, hotels, and parking structures and are often multifunctional areas that can be used for many activities including gardening, relaxing, and socializing. To enable activities for people and the integration of larger plants, trees, and architectural elements on green roofs, suitable rooftops need to be relatively flat and fulfil more complex technical requirements e.g., regarding weight.



4.4 Smart roof

Smart roofs are a unique type of intensive or extensive green roof that provide several services to protect ecosystems in cities. Many of the benefits are similar to other green roofs (e.g., basic habitat provision, reduction of localized air temperature, stormwater management). However, capillary smart roofs represent an extension of conventional green roofs because the system is equipped with a drainage system under the vegetation layer. The drainage layer retains stormwater, thereby reducing flood risk more so than a typical green roof. Through capillary fiber cylinders, water is naturally returned to the vegetation layer during dry periods. Capillary smart roofs represent cyclic water management where additional plant irrigation is not needed (100% of the stormwater can be reused for irrigation).



4.5 Green façades

Planted façades with controlled cultivation are called green façades. Green façades offer many benefits including reduction of air pollution, thermal insulation for buildings, and biodiversity support via habitat provision and connectivity improvement. Façade greenings are divided into two types: façade-bound greening and ground-based greening.

Façade-bound greening uses panels or containers that are fixed to the façade or is part of the façade itself. Vegetation is usually planted directly in the thin substrate of the panel and then elevated. Therefore, façade-bound greening systems do not rely on climbing plants, and can be removed during winter.

Ground-based green façades are established using climbing plants. The climbing vegetation is planted in the ground and therefore extracts water and nutrients directly from the soil. The vegetation grows directly on the wall, or climbs on a frame that is connected to, but keeps a small distance from the wall.



2.1 Single line street trees

Single line street trees represent one possibility to establish several trees in urban areas. As the name implies, single line trees are arranged along one side of streets, bicycle paths, sidewalks, or other pathways.

Trees in general can positively affect local microclimate conditions, absorb gaseous pollutants, intercept particulate matter, and provide shade for people and buildings. One of the main positive effects for human well-being in warmer periods is the mitigation of urban heat stress due to shading and plant transpiration. The potential effects of street trees depend on factors such as tree size, canopy cover, planting density, species, tree health, location, availability of root water, and leaf area index.



2.2 Boulevards

Boulevards represent a possibility to establish several trees in cities to mitigate urban heat stress, while providing additional benefits like improving water management and climate resilience. Within boulevards, trees are commonly arranged along streets, bicycle paths, and sidewalks on both sides of the road. The canopies of opposite trees often form a (nearly) closed canopy. As a result, the area between the two tree lines is shaded and the air temperature cools.



2.3 Group of trees

Groups of trees mimic the gestalt of a forest in an urban setting. They may be an option for the design of shaded squares, as a contrasting element in densely built areas, or for courtyard design. In some urban areas, groups of trees may also be developed from existing, wild growing trees that established spontaneously and are typical pioneer species of urban forests. Urban groups of trees offer many benefits like improved water management and climate resilience and contribute to the mitigation of urban heat stress. Additionally, selection of diverse native species, especially in combination with understory vegetation, can help support and enhance biodiversity.



4.6 Free standing living wall

Verticalization of green spaces is a method to increase vegetated surfaces with many ecological services in urban environments. Free-standing living walls serve as adaptation measures for the urban heat island effect. Furthermore, they create space with high amenity value and potentially high biodiversity. Free-standing living walls can also be used as noise barriers along highly frequented roads. They are suitable to re-use stormwater runoff water and have a high rate of evapotranspiration.

International Union for Conservation of Nature. (2020). *Global standard for nature-based solutions: A user-friendly framework for the verification, design and scaling up of NbS* (1st ed.). IUCN.

<https://doi.org/10.2305/IUCN.C.H.2020.08.en>



Fig 4.9 Green façade with a façade-bound greening in Reutlingen, Germany.

Green facades

- Limited space
- Increased insulation
- Reduces temperature and air pollution



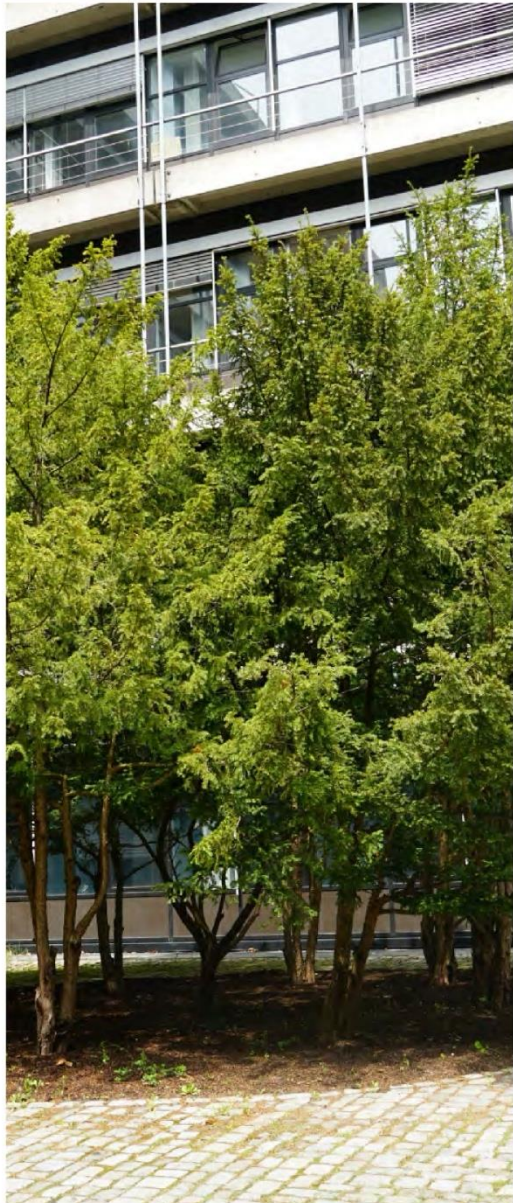


Fig 4.12 Free standing living wall in Ludwigsburg, Germany.

Free standing living wall

- Some space
- Increases vegetation without restructuring streets
- Reduces temperature and air pollution





Groups of trees

- More space
- Shading + evapotranspiration increases comfort
- Reduces temperature and air pollution + space quality

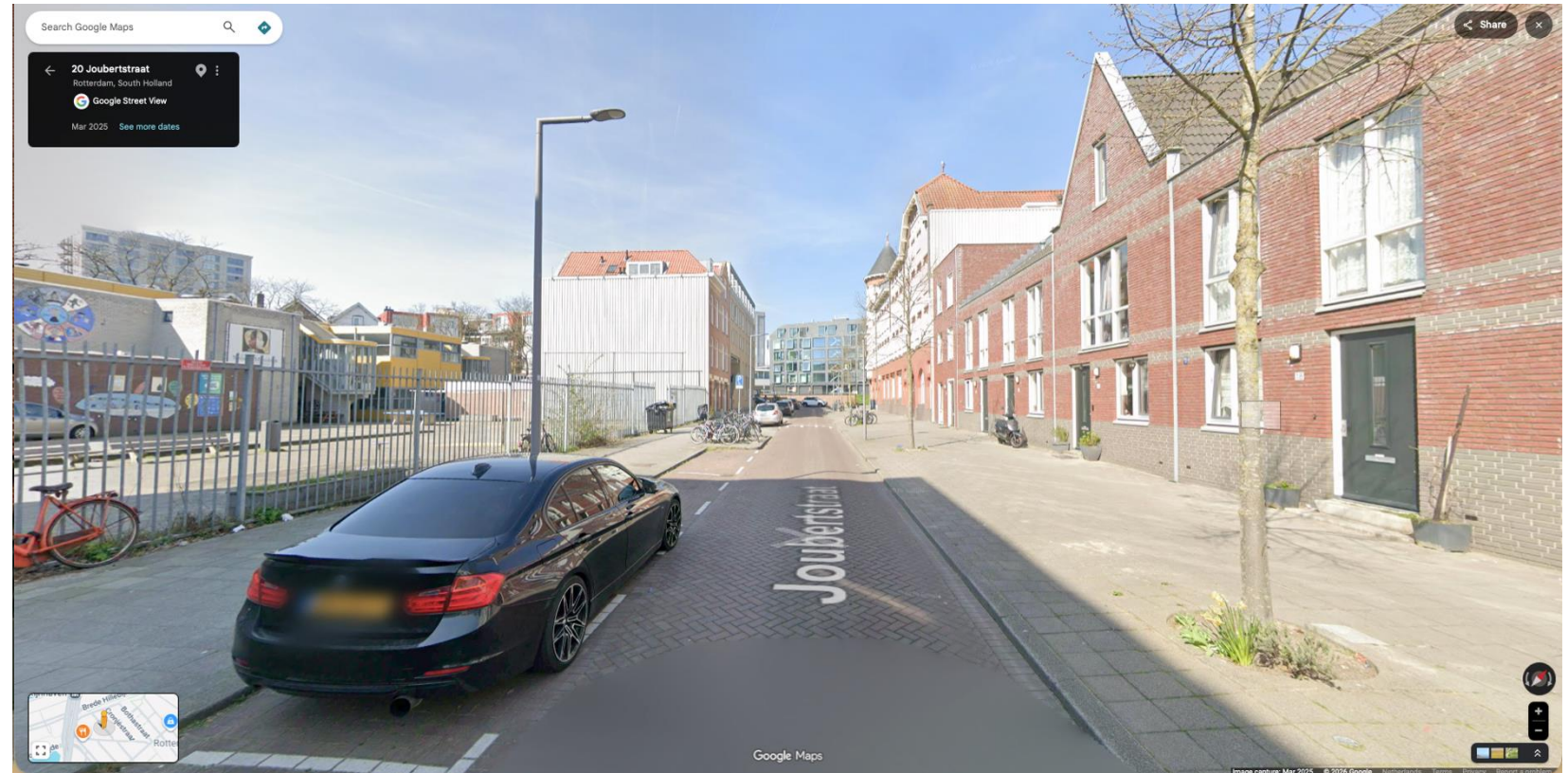


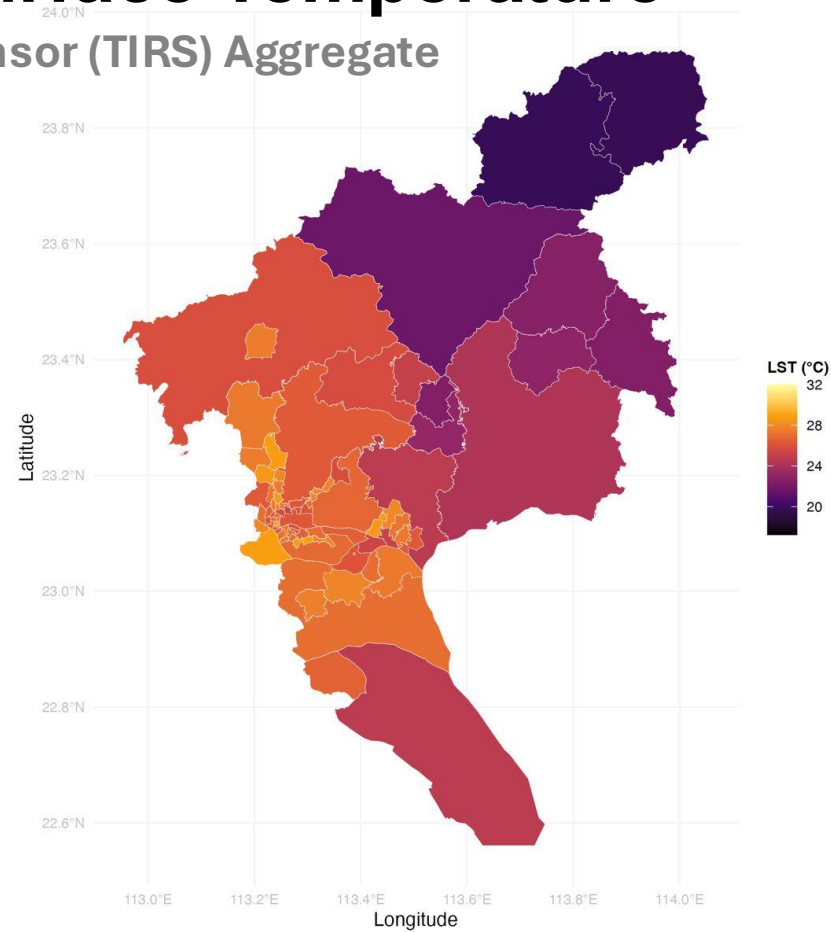
Fig 2.4 Group of trees in a courtyard in Stuttgart, Germany.

05

Workflow Reproduction: Guangzhou

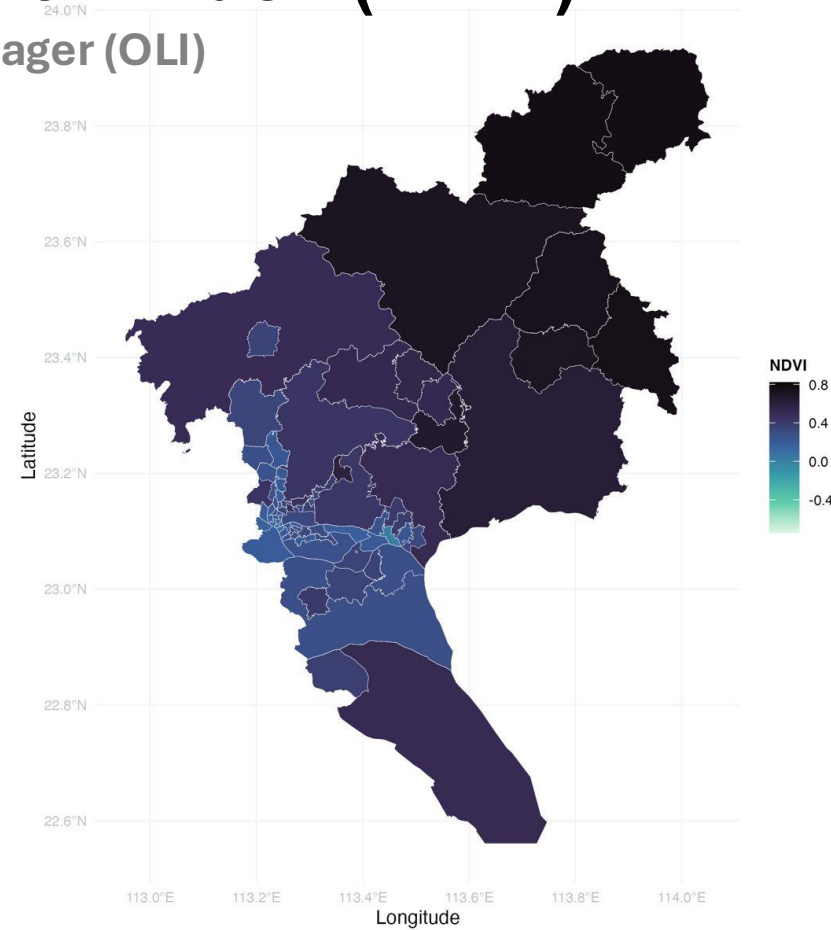
Guangzhou Land surface Temperature

Lansat 8/9 Thermal Infrared Sensor (TIRS) Aggregate



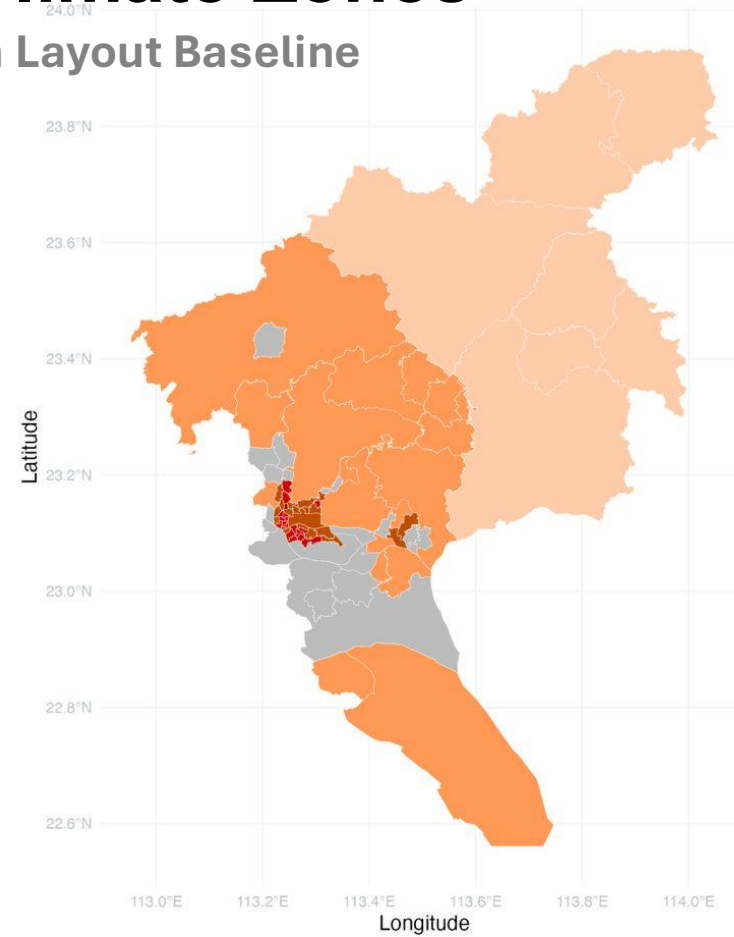
Guangzhou Vegetation Index (NDVI)

Lansat 8/9 Operational Land Imager (OLI)



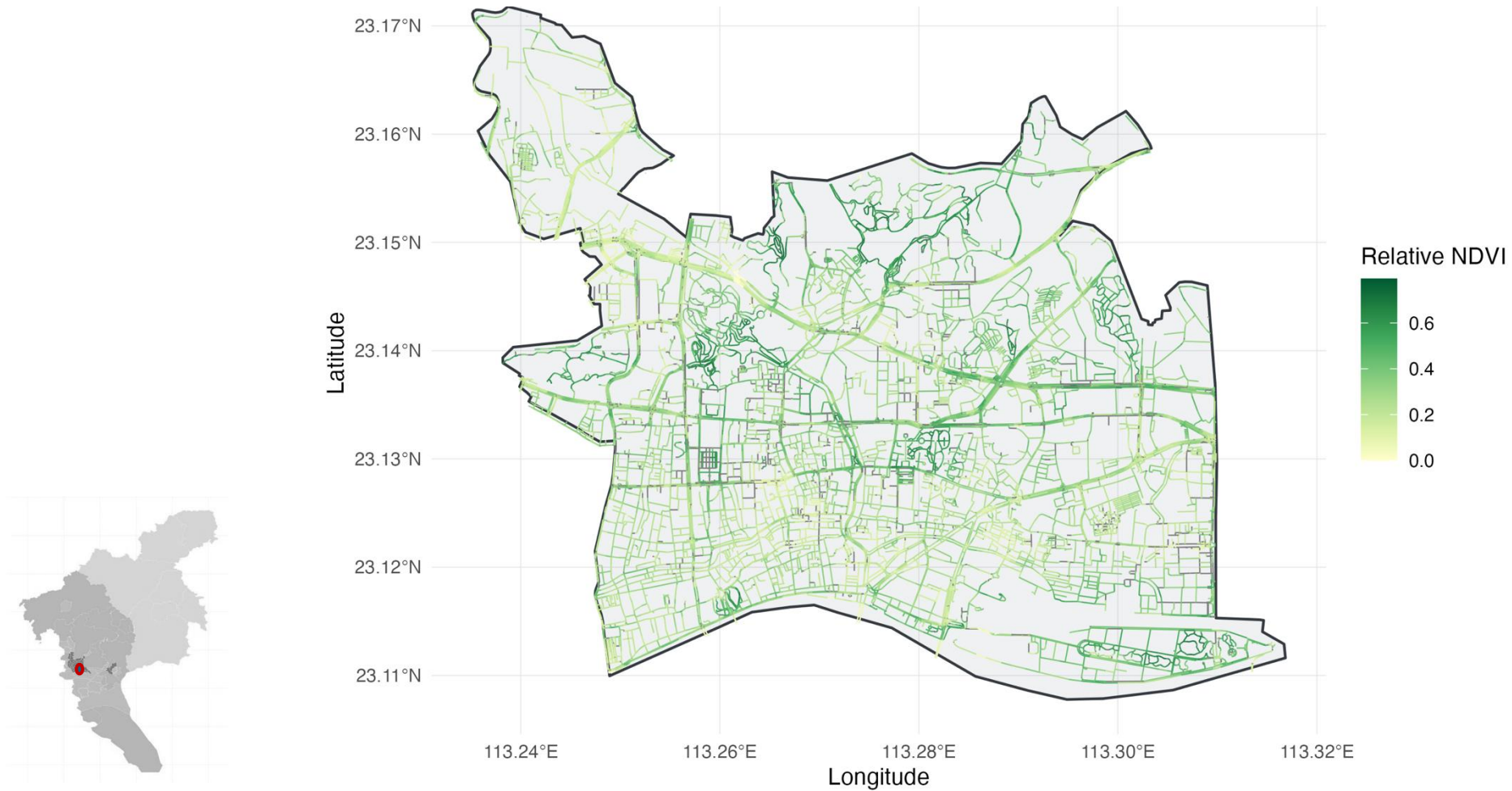
Guangzhou Local Climate Zones

RUBCLIM Morphological Urban Layout Baseline



Street-Level Vegetation Analysis (NDVI)

Yuexiu District (Guangzhou)



Street-Level Land Surface Analysis (LST)

Yuexiu District (Guangzhou)

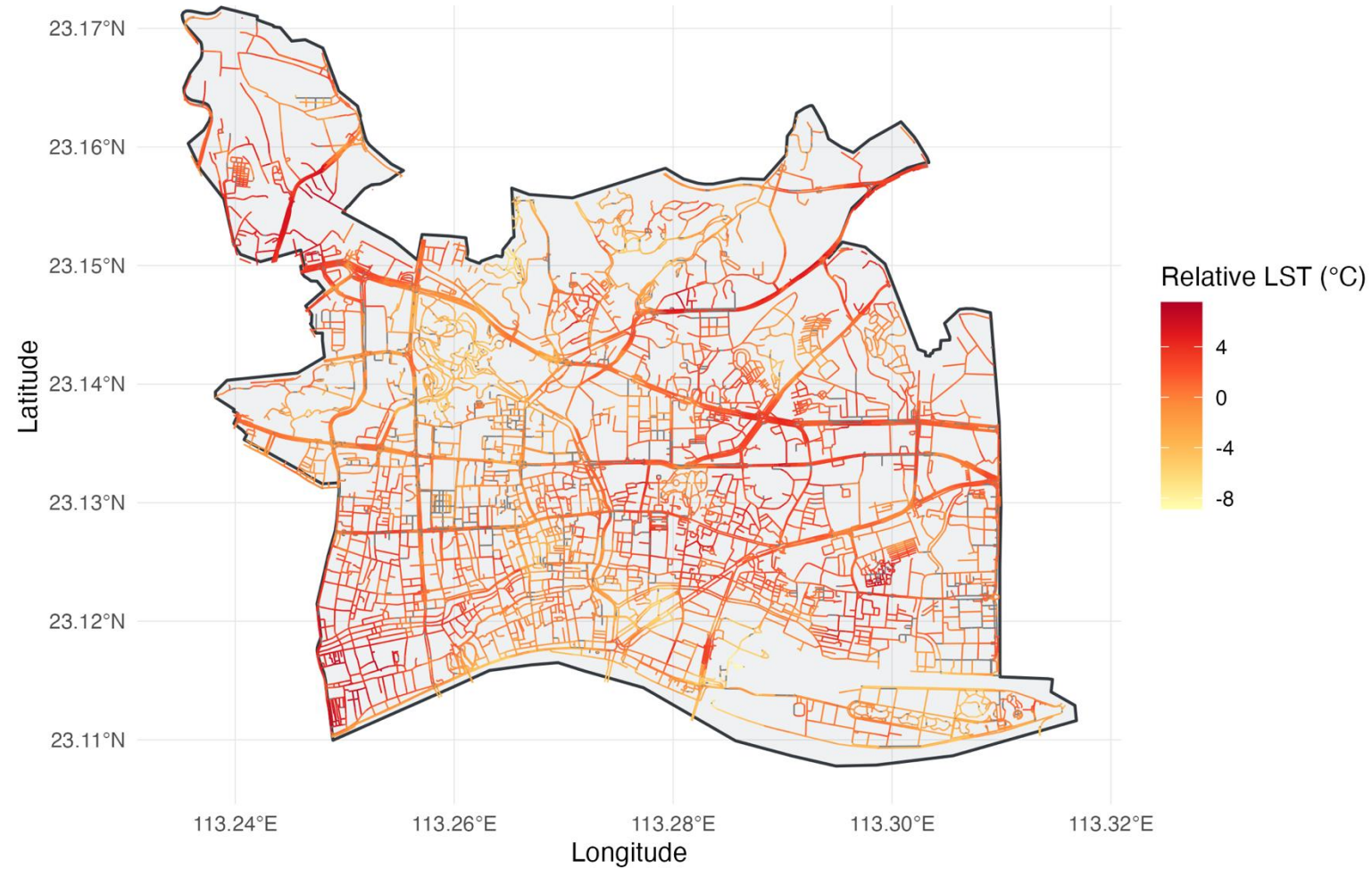




Fig A.9 Green facade with a facade-bound greening in Reutlingen, Germany.

Green facades

- Little additional space
- Can be implemented into existing structures
- Reduces temperature and, cooling stress

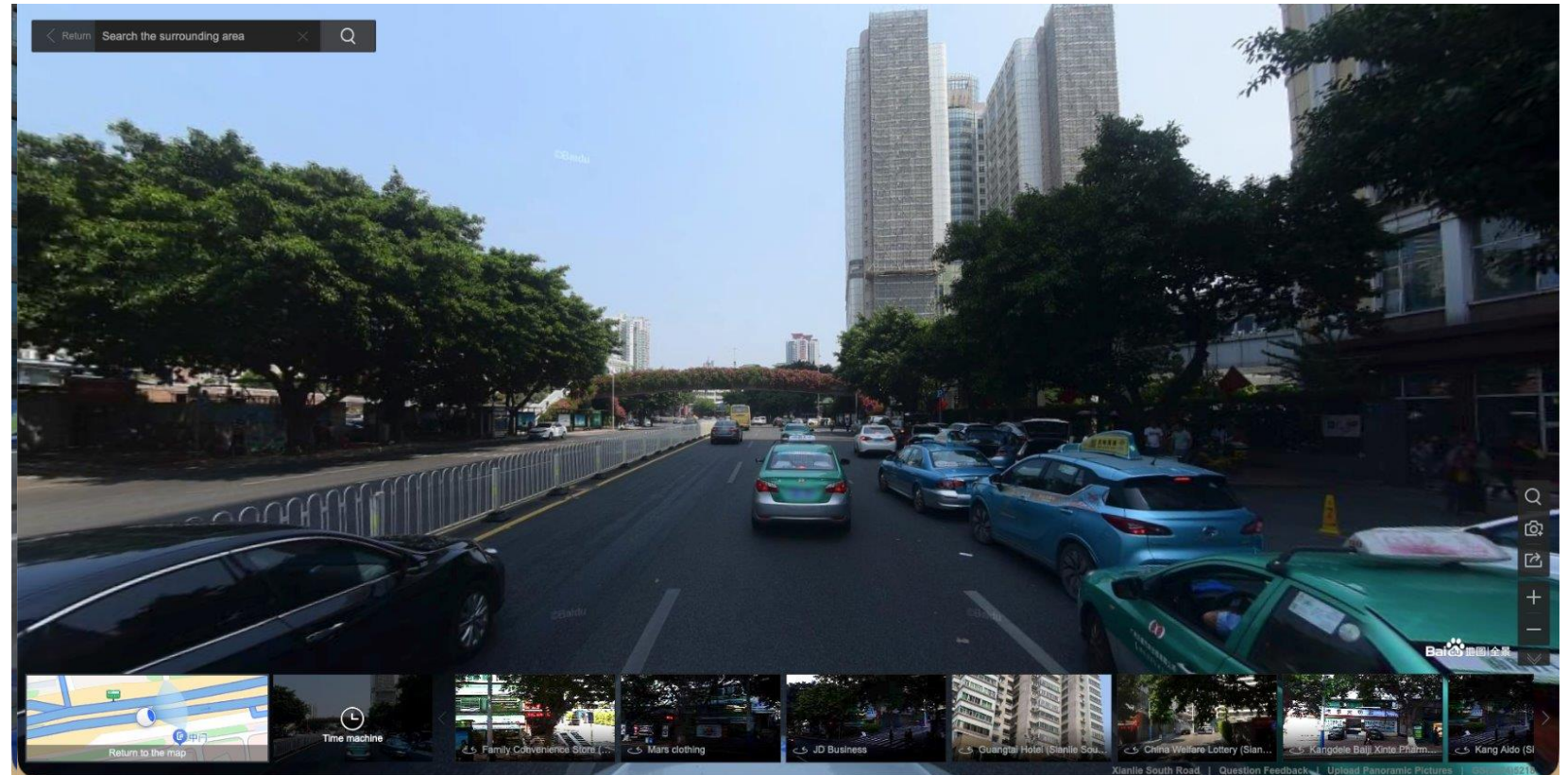
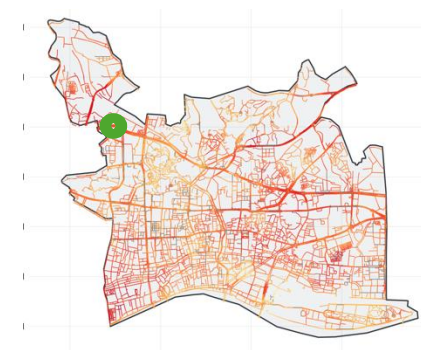




Fig 4.12 Free standing living wall in Ludwigsburg, Germany.

Green walls

- Some space
- Sound comfort
- Reduces temperature and air pollution





Groups of trees

- More space
- Increases shading and evapotranspiration
- Reduces temperature and air pollution

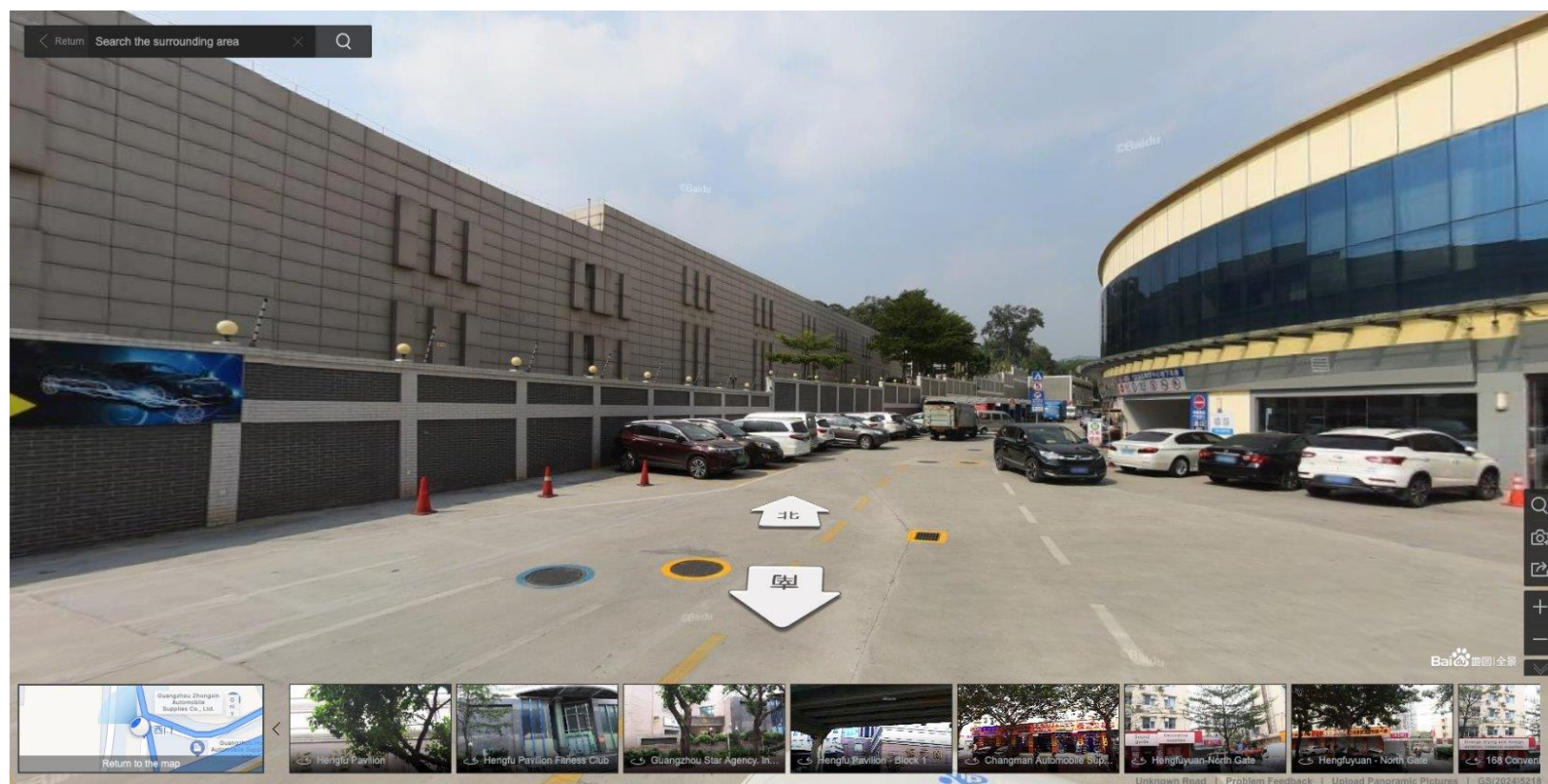


Fig 2.4 Group of trees in a courtyard in Stuttgart, Germany.

06

Discussion

How do spatial scales influence analysis of thermal impacts?

Spatial scale matters more than expected

- neighbourhood maps show broad thermal patterns across Rotterdam
 - warmer industrial and dense urban areas versus greener areas
- City scale averages smooth local hotspots
- Actionable interventions happen at street scale

Vegetation consistently corresponds to lower temperatures

- Thanks to evapotranspiration and shading
- Wind speeds, humidity were not included in the analysis
- Actionable interventions happen at street scale

07

Conclusion

The scale of analysis fundamentally changes what interventions become visible.

- Satellite remote sensing offers a valuable macro-level perspective of surface temperatures, but its data accuracy and temporal resolution limit its utility for detailed urban planning
- While neighbourhood-level aggregation is useful for identifying broad patterns, street-level analysis revealed substantial intra-neighbourhood variability that would otherwise remain hidden.

UGI Benefits & Impacts

- can **lower LST** by 28 °C
- Highest temperatures occur in less accessible parts of the city

Urban context matters

- Limited space
- High building density
- Little space for large green infrastructures

Thank you for your attention!